

Frequent Item & fp growth

* Find the most frequent item with min-support = 3

TID	Items
T1	Coffee, Milk, Sugar
T2	Coffee, Milk
T3	Coffee, Sugar
T4	Milk, Sugar, Tea
T5	Coffee, Milk, Sugar
T6	Coffee, Tea

Support

$$(A \rightarrow B) = \frac{\text{count}(A \cap B)}{\text{count}(A)}$$

[Percentage]

Confidence

$$(A \rightarrow B) = \frac{\text{count}(A \cap B)}{\text{count}(A)}$$

$L_1 \rightarrow C_1$

item	Count
Coffee	5
Milk	4
Sugar	4
Tea	2

✓

✓

✓

X Rejected

لا زيم ال count
يكون اكبر من
او يساوي ال
min-support

* ال Tea Rejected فمش هيروح مخايا ال itemset
اللي بعده

$L_2 \rightarrow C_2$

items	Count
{Coffee, Milk}	3
{Coffee, Sugar}	3
{Milk, Sugar}	3

$L_3 \rightarrow C_3$

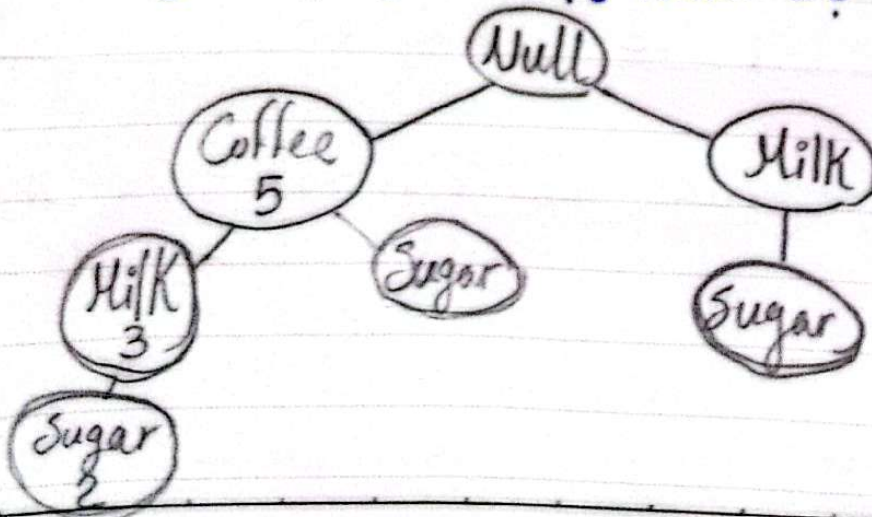
items	Count
{Coffee, Milk, Sugar}	2

→ Rejected

* هحدد ال frequent item من افر جدول كان فيه Accepted
 ← هنا: تاف جدول

* fp growth

← لازم اعمل اول itemset واشيل ال Rejected
 ← ببدأ دائما بال Null وبعدها احتراجة ظهرت



Transaction Id	Items
1	A, B, E
2	A, B, D, E
3	B, C, D, E
4	B, D, E
5	A, B, D
6	B, E
7	A, E

count ≥ 3

$L_1 \rightarrow C_2$

Item	Count
A	4
B	6
C	1
D	4
E	6

$\Rightarrow$ Rejected

$L_2 \rightarrow C_2$

Items	Count
{A, B}	3
{A, D}	2
{A, E}	3
{B, D}	4
{B, E}	5
{D, E}	3

$\Rightarrow$ Rejected

$L_3 \rightarrow C_3$

Items	Count
{A, B, E}	2
{B, D, E}	3

$\Rightarrow$ Rejected

* Support of the itemset {B, D, E} :

$$\text{Support} = \frac{\text{Count}(\{B, D, E\})}{DB} = \frac{3}{7} \approx 43\%$$

* Support & confidence of the association rule $BD \rightarrow E$:

$$\rightarrow \text{Support}(BD \rightarrow E) = \frac{\text{Count}(BD \cap E)}{DB} = \frac{3}{7} \approx 43\%$$

$$\begin{aligned} \rightarrow \text{Confidence}(BD \rightarrow E) &= \frac{\text{Count}(BD \cap E)}{\text{Count}(BD)} \\ &= \frac{3}{4} \end{aligned}$$